

WEEK-2 Assignment

Task Name: Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Description

The objective of this task is to design and implement an Intelligent Inventory Replenishment System using Reinforcement Learning techniques. The system simulates how businesses automatically manage inventory levels based on customer demand, supplier lead times, and stock availability while minimizing inventory costs and preventing stockouts.

Each inventory replenishment decision is treated as an action, where the system learns the optimal ordering policy through continuous interaction with the environment. Demand patterns, holding costs, and shortage costs are considered to evaluate the effectiveness of each decision. The reinforcement learning agent continuously updates its policy to maximize long-term rewards by balancing inventory availability and operational costs.

The engine also adapts to changing business conditions such as seasonal demand fluctuations, supplier delays, unexpected demand spikes, and promotional events, ensuring efficient inventory management over time.

Objectives:

Design an intelligent inventory replenishment simulation.

Model inventory decisions using reinforcement learning.

Analyze customer demand and stock availability.

Learn optimal ordering policies through continuous feedback.

Balance inventory holding costs and stockout risks.

Adapt to seasonal demand and supplier disruptions.

Evaluate system performance using inventory cost, service level, and stock availability metrics.

Expected Outcome:

Students will develop an intelligent inventory replenishment system capable of learning from demand patterns, adapting to changing market conditions, optimizing stock levels, and improving overall supply chain efficiency using reinforcement learning techniques.